

- 6 (a) The magnetic flux density of a magnetic field is numerically equal to the force per unit length acting on a long straight conductor carrying a unit current at right angles to the magnetic field.

- (b) (i) Since the α -particle is positively charged, electric force is acting downwards.

The magnetic force acting on the α -particle must be upwards for it to be traveling undeflected. By Fleming's Left hand rule, the magnetic field is pointing into the page.

(ii) $F = qE = (2)(1.60 \times 10^{-19})(500) = 1.60 \times 10^{-16} \text{ N}$

(iii)

$$KE = \frac{1}{2}mv^2$$

$$120 \times 1.60 \times 10^{-19} = \frac{1}{2}(6.64 \times 10^{-27})v^2$$

$$v = 7.60 \times 10^4 \text{ m s}^{-1}$$

(iv) $qE = Bqv$

$$E = Bv$$

$$500 = B(7.60 \times 10^4)$$

$$B = 6.58 \times 10^{-3} \text{ T}$$

- (v) Since the mass of the proton is smaller than that of the α -particle, the velocity of the proton will be greater than that of the α -particle.

Both forces are proportional to charge. However, the magnetic force is also proportional to velocity. The magnetic force acting on the proton is greater than electric force.

Proton deflects upwards towards plate P.